NBTS

Aakash + BYJU'S

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MM : 720 **Time : 3.00 Hrs**

NCERT Booster Test Series

(*for* NEET-2022)

Test - 7

Topics covered :

Physics : Electric Charges and Fields, Electrostatic Potential and Capacitance.
Chemistry : The Solid State, Solutions, Electrochemistry, Chemical Kinetics.
Botany : Reproduction in Organisms, Sexual Reproduction in Flowering Plants.
Zoology : Reproduction in Organisms, Human Reproduction.

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
(iii) Use blue/black ballpoint pen only to darken the appropriate circle.
(iv) Mark should be dark and completely fill the circle.
(v) Dark only one circle for each entry.
(vi) Dark the circle in the space provided only.
(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

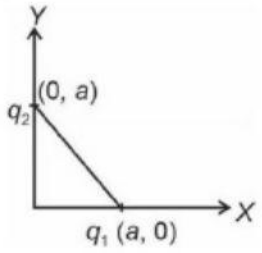
PHYSICS

Choose the correct answer :

SECTION-A

1. When some charge is transferred to a conductor.
   (1) It readily gets distributed over the entire volume of the conductor
   (2) It readily gets distributed over the entire surface of the conductor
   (3) It stays at the same place and does not distribute over the surfaces or in volume of the conductor
   (4) Its potential must increase

2. A positive point charge q_1 is placed at $(a, 0)$ and another positive point charge q_2 is placed at $(0, a)$ as shown in the figure. Find the force acting on q_1?



BOTANY

SECTION-A

101. With respect to the life span of various organisms, choose the **correct** representation.
(1) Crow > Crocodile
(2) Banana plant > Rice plant
(3) Butterfly > Rose plant
(4) Fruitfly > Crow

102. Cell division is itself a mode of reproduction in
(1) Monerans and protists
(2) All plants and most of the animals
(3) Eukaryotes only
(4) All living organisms

103. Select the characteristic(s) which is/are **true** for both the plants, bamboo and neelakurinji.
(a) They are monocots
(b) They flower only once in their life
(c) These are perennial plants
(d) They have long interflowering periods
(1) (b), (c) and (d) (2) (b) and (c) only
(3) (b) only (4) (a) and (b)

104. Which process enables the continuity of the species, generation after generation?
(1) Growth (2) Hybridisation
(3) Gametogenesis (4) Reproduction

105. Morphologically and genetically similar individuals are usually formed by all of the following ways, **except**
(1) Budding (2) Fragmentation
(3) Fusion of gametes (4) Binary fission

106. The phenomenon of encystation is seen
(1) In some unicellular organisms
(2) Primarily under favourable conditions
(3) Only in those organism that have cell wall
(4) Only in prokaryotes

107. Vegetative propagule in ginger is
(1) Offset (2) Tuber
(3) Rhizome (4) Bulb

108. Amongst the following plants, shortest juvenile phase is of
(1) Mango (2) Rice
(3) Bamboo (4) Coconut

109. Read the following statements and choose the option which is **true** for them.

Statement-1: In plants, hormones are responsible for the transitions between different phases of growth.

Statement-2: Environmental factors do not play any role in the regulation of reproductive process in plants.
(1) Only statement-1 is correct
(2) Only statement-2 is correct
(3) Both the statements are correct
(4) Both the statements are incorrect

110. Which of the following are main pre-fertilisation events?
(a) Triple fusion (b) Syngamy
(c) Gametogenesis (d) Gamete transfer
(1) (a) and (c) (2) (c) and (d)
(3) (b) and (c) (4) (a) and (d)

111. In majority of sexually reproducing organisms
(1) Male and female gametes are morphologically similar
(2) Male gametes are larger than female gametes
(3) Male gametes are called antherozoids or sperms
(4) Male gametes are non-motile

112. In several fungi, the term homothallic is used. It indicates that
(1) They form two different types of gametes
(2) They form same type of gametes
(3) These fungi are formed of same type of mycelia
(4) Both male and female gametes are produced by the same thallus

113. Gametes are __A__ and the parent plant body from which they are produced __B__.

Select the **correct** option to fill in the blanks 'A' and 'B'.

	A	B
(1)	Always haploid	May be haploid or diploid
(2)	Haploid or diploid	Are always diploid
(3)	Haploid	Are always diploid
(4)	Diploid	Are always diploid

114. In which of the following structures, three wall layers of protection is found?

(1) Archegonium of pteridophytes
(2) Antheridium of bryophytes
(3) Anther of typical stamen
(4) Seed of gymnosperm

115. Match the following columns and select the **correct** option.

	Column I		Column II
A.	Pollen grain	(i)	Locule
B.	Ovarian cavity	(ii)	Germ pores
C.	Ovule	(iii)	Filiform apparatus
D.	Synergid	(iv)	Micropyle

	A	B	C	D
(1)	(iii)	(iv)	(ii)	(i)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(iii)	(i)	(ii)	(iv)

116. Plants such as *Commelina* can perform

(a) Autogamy (b) Geitonogamy
(c) Xenogamy

The correct one(s) is/are

(1) (a) only
(2) (a) and (b) only
(3) (c) only
(4) All (a), (b) and (c)

117. Wind pollination is quite common in

(1) Members of family Fabaceae
(2) Grasses
(3) Flowering plants that produce sticky pollen grains
(4) Dicotyledonous plants only

118. The hollow foliar structure that encloses shoot apex and leaf primordia in grass embryo is

(1) Coleorrhiza (2) Plumule
(3) Coleoptile (4) Scutellum

119. Choose the **incorrect** statement from the following w.r.t. a typical angiosperm anther.

(1) It is bilobed
(2) Each lobe has two theca
(3) Each microsporangium develops into two pollen sacs
(4) It consists of four microsporangia

120. In angiosperms, ploidy of cells of microspore tetrad is

(1) 2n
(2) n
(3) 4n
(4) 3n

121. Pollen grain in flowering plants represents

(1) Its female gametophyte
(2) Gametophyte generation
(3) A part of sporophyte generation
(4) Microspore forming structure

122. The organic material that is **not** degraded by any enzyme so far known is found in

(1) Exine of pollen grain
(2) Tapetum of anther wall
(3) Intine of pollen grain
(4) Endothecium of anther wall

123. A mature male gametophyte in angiosperm is _____ structure.

Select the **correct** option to fill in the blank.

(1) Multinucleate (2) Three-celled
(3) Dikaryotic (4) Two-celled

124. Gynoecium of *Michelia* is
(1) Apocarpous (2) Monocarpellary
(3) Syncarpous (4) Bicarpellary

125. Select the **incorrect** match from the following.
(1) Chalaza – Basal part of ovule
(2) Hilum – Encloses the female gametophyte
(3) Funicle – Stalk of ovule
(4) Nucellus – Has reserve food material

126. Embryo sac in flowering plants is
(1) Directly attached to the placenta
(2) Formed by the cells of chalaza
(3) Located at chalazal pole of ovule
(4) Formed from the cell of nucellus

127. To achieve fertilisation, the phenomenon of pollination occurs in
(1) Seed producing dioecious plants only
(2) All monoecious and dioecious plants
(3) All angiosperms and gymnosperms
(4) All the plants that produce non-motile male gametes

128. In which of the following plants, only chasmogamous flowers are present?
(1) *Oxalis*
(2) China rose
(3) Common pansy
(4) *Commelina*

129. For pollination, majority of flowering plants use
(1) Wind
(2) Gravity
(3) Biotic agents
(4) Water

130. Both wind and water pollinated flowers
(1) Are very colourful
(2) Remain closed at maturity
(3) Do not produce nectar
(4) Produce very less amount of pollen grains

131. Flowering plants have developed many devices to discourage self-pollination. Which one of the following is **not** amongst them?
(1) No synchronisation between pollen release and stigma receptivity
(2) Different position of anther and stigma within a flower
(3) Inhibition of pollen germination on the stigma of flower of the same plant
(4) Bearing male and female flowers on the same plant

132. Which of the following phenomena is **not** considered as an event of pollen-pistil interaction?
(1) Deposition of pollen on the stigma
(2) Entry of pollen tube into the ovule
(3) Germination of pollen grain on the stigma
(4) Release of pollen from anther

133. Match the following columns and select the **correct** option.

	Column I		**Column II**
A.	Black pepper	(i)	False fruit
B.	Mango	(ii)	Integument of ovule
C.	Strawberry	(iii)	Perisperm
D.	Seed coat	(iv)	Edible fleshy mesocarp

A B C D
(1) (ii) (iii) (iv) (i)
(2) (iii) (iv) (i) (ii)
(3) (iv) (i) (iii) (ii)
(4) (iii) (ii) (i) (iv)

134. Select the **incorrect** statement w.r.t. cotyledons.
(1) It is made up of diploid cells
(2) It is a part of embryo
(3) It is absent in monocotyledonous embryos
(4) It stores food materials

135. Epiblast is a part of
(1) Outer covering of a fruit
(2) Embryo of some grasses
(3) Anatropous ovule
(4) Dicot seed

SECTION – B

136. Choose the **incorrect** match from the following.
(1) Water hyacinth – Offset
(2) Penicillium – Conidia
(3) Bryophyllum – Zoospores
(4) Agave – Bulbil

137. Syngamy does **not** occur inside the body of organism in
(1) Cycas (2) Funaria
(3) Dryopteris (4) Ulothrix

138. The phenomenon of spore formation does **not** occur in
(1) Protists (2) Plants
(3) Vertebrates (4) Monerans

139. Nodes and internodes are **not** present in
(1) Rhizome (2) Tuber
(3) Root (4) Offset

140. Select the **incorrect** statement from the following.
(1) A meiocyte of maize has more number of chromosomes as compared to that of rice
(2) Male sex organ of *Chara* is called antheridium
(3) *Marchantia* is a dioecious plant
(4) Papaya plant has either staminate flowers or pistillate flowers

141. In a typical embryo sac, the central cell has
(1) Two polar nuclei
(2) One polar nuclei and one egg nucleus
(3) Two synergids and one egg nucleus
(4) Only one polar nucleus

142. Select the **correct** statement regarding apomixis.
(1) It is a form of asexual reproduction that mimics sexual reproduction
(2) Seeds so formed do not have embryos
(3) This process occurs only in hybrid varieties
(4) In this process, segregation and reunion of chromosomes occur

143. Mature seeds are ex-albuminous in
(1) Maize (2) Groundnut
(3) Wheat (4) Castor

144. All of the following are the characteristics of the generative cell of a pollen grain, **except**
(1) Larger than vegetative cell
(2) Floats in the cytoplasm of the vegetative cell
(3) Spindle shaped
(4) Dense cytoplasm

145. In majority of flowering plants
(1) Male gametes are motile
(2) One megaspore develops into four embryo sacs
(3) Embryo sac development is monosporic
(4) Embryo sac is polyploid

146. In which of the following plants, only xenogamy is possible?
(1) Papaya (2) Maize
(3) Mustard (4) Cucurbit

147. In *Zostera*, pollination occurs by
(1) Animals
(2) Water
(3) Wind
(4) Both wind and insects

148. The reward(s) for pollinating animals from flowers can be
(a) Fragrance
(b) Nectar
(c) Pollen grains
The **correct** one(s) is/are
(1) All (a), (b) and (c) (2) Only (b) and (c)
(3) (b) only (4) (c) only

149. Dioecy is seen in
(1) Pea (2) Castor
(3) Rose (4) Date palm

150. After fertilisation, what happened with antipodal cells?
(1) One of them participates in triple fusion and others degenerate
(2) All of them degenerate
(3) Two of them participate in triple fusion and one degenerates
(4) All of them multiply to form a tissue that stores food